

```
import java.util.Scanner;
public class Quiz02SetA {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        double total = 0.0;
        while (true) {
            System.out.print("Enter item name (or 'done' to stop): ");
            String name = sc.nextLine();
            if (name.equals("done")) {
                break; // stop taking items
            }
            System.out.print("Enter price: ");
            double price = sc.nextDouble();
            sc.nextLine(); // clear input buffer
            if (price <= 0) {
                System.out.println("Invalid price. Item ignored.");
                continue; // skip invalid item
            }
            total += price;
        }
        System.out.print("Membership card? (true/false): ");
        boolean hasCard = sc.nextBoolean();
        double discountPercent = 0.0;
        double finalAmount = total;
        if (total > 5000) {
            double firstDiscount = finalAmount * 0.10; // 10% discount
            finalAmount -= firstDiscount;

            if (hasCard) {
                double secondDiscount = finalAmount * 0.05; // extra 5% on reduced amount
                finalAmount -= secondDiscount;
                discountPercent = 100 * (1 - (finalAmount / total));
            } else {
                discountPercent = 10.0;
            }
        }

        System.out.println("Total before discount: " + total + " taka");
        System.out.printf("Discount applied: %.1f%%\n", discountPercent);
        System.out.println("Final amount: " + finalAmount + " taka");
    }
}
```

Mrs. Karim is preparing for her weekly grocery shopping and wants to calculate her final bill after applying discounts. She plans to buy multiple items but wants to stop whenever she types "**done**" instead of an item name.

The store rules are as follows:

- Each item has a **name** and a **price**.
- If the **price is less than or equal to zero**, that item is **ignored** from the total.
- If she enters "**done**", the input ends immediately.
- After entering all items, she specifies whether she has a **membership card** (true or false).

Discount policy:

- If her total spending exceeds **5000 taka**, she gets a **10% discount**.
- If she also has a **membership card**, she gets an **extra 5% discount on the discounted amount after the first discount is applied**.

Write a program that calculates and prints:

1. Total before discount
2. Total discount percentage on the initial bill
3. Final payable amount after all discounts

Sample Input & Output with Explanation

Input	Output
Enter item name (or 'done' to stop): Rice Enter price: 3200 Enter item name (or 'done' to stop): Oil Enter price: 2500 Enter item name (or 'done' to stop): done Membership card? true	Total before discount: 5700 taka Discount applied: 14.5% Final amount: 4873.5 taka
Enter item name (or 'done' to stop): Fruits Enter price: -200 Enter item name (or 'done' to stop): Milk Enter price: 0 Enter item name (or 'done' to stop): Bread Enter price: 180 Enter item name (or 'done' to stop): done Membership card? false	Total before discount: 180 taka Discount applied: 0% Final amount: 180 taka

```
import java.util.Scanner;
public class Quiz02SetB {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        double total = 0.0;
        while (true) {
            System.out.print("Enter book title (or 'end' to stop): ");
            String title = sc.nextLine();
            if (title.equals("end")) {
                break; // stop taking inputs
            }
            System.out.print("Enter price: ");
            double price = sc.nextDouble();
            sc.nextLine(); // clear buffer
            if (price <= 0) {
                System.out.println("Invalid price. Book ignored.");
                continue; // skip this entry
            }
            total += price;
        }
        System.out.print("Loyalty member? (true/false): ");
        boolean isMember = sc.nextBoolean();
        double delivery = 100.0;
        if (total > 4000.0) {
            delivery = 0.0; // waived
        }
        double rewardPoints = 0.0;
        if (isMember) {
            rewardPoints = total * 0.02; // 2% of total (before delivery)
        }
        double finalAmount = total + delivery;
        System.out.printf("Total book cost: %.1f taka%n", total);
        System.out.printf("Delivery charge: %.1f taka%n", delivery);
        System.out.printf("Reward points: %.1f%n", rewardPoints);
        System.out.printf("Final amount: %.1f taka%n", finalAmount);
    }
}
```

Question 01 [10 Marks]

Mr. Hassan runs an online bookshop. At the end of each day, he wants to calculate how much a customer has to pay after adding delivery charges and applying loyalty rewards. He keeps entering the **titles** and **prices** of books purchased by a customer. When he types "**end**", the entry stops.

The billing rules are as follows:

- Each book has a **title** and a **price**.
- If the **price is less than or equal to zero**, that book is **ignored**.
- A **delivery charge of 100 taka** is added to every order.
- If the **total purchase (before delivery)** exceeds 4000 taka, the delivery charge is **waived**.
- If the customer is a **loyalty member**, they earn a **reward point equal to 2%** of the total purchase (before delivery).

Your program should display:

1. Total book cost (before delivery)
2. Delivery charge applied
3. Reward points earned (if any)
4. Final payable amount

Sample Input & Output with Explanation

Input	Output
Enter book title (or 'end' to stop): Java Basics Enter price: 2500 Enter book title (or 'end' to stop): Data Science 101 Enter price: 1900 Enter book title (or 'end' to stop): end Loyalty member? true	Total book cost: 4400.0 taka Delivery charge: 0.0 taka Reward points: 88.0 Final amount: 4400.0 taka
Enter book title (or 'end' to stop): Algorithms Enter price: 3500 Enter book title (or 'end' to stop): C Programming Enter price: -200 Enter book title (or 'end' to stop): end Loyalty member? false	Total book cost: 3500.0 taka Delivery charge: 100.0 taka Reward points: 0.0 Final amount: 3600.0 taka

Brac University
Department of Computer Science and Engineering

CSE110: PROGRAMMING LANGUAGE I Fall 2025 Quiz 3 Marks: 10

Name:	ID:	Section:
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Question 1: You are a novice programmer tasked with creating a magical character pyramid for a wise wizard. The wizard has given you a single uppercase letter from the English alphabet (A-Z) and has asked you to generate a pyramid of characters. Each row of the pyramid grows longer, starting from the first letter of the alphabet ('A') and continuing up to the given letter.

The wizard's challenge is simple yet enchanting:

- The first row will have only the letter 'A'.
- The second row will include the first two letters of the alphabet ('A', 'B').
- This pattern will continue until the final row includes all the letters from 'A' to the wizard's chosen letter.

Your task is to impress the wizard with your programming skills by generating the pyramid dynamically based on the letter he provides. Good luck on this magical quest!

Sample Input	Sample Output
E	A AB ABC ABCD ABCDE
C	A AB ABC
A	A

Tentative Solution:

```
import java.util.Scanner;

public class CharacterPyramid {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter an uppercase character (A-Z): ");

        char input = scanner.next().charAt(0);

        for (char i = 'A'; i <= input; i++) {

            for (char j = 'A'; j <= i; j++) {

                System.out.print(j);

            }

            System.out.println();

        }

    }

}
```

Question 2: Trace the following code and generate the outputs. [Bonus: 3 Marks]

1	<code>public class Q10 {</code>
2	<code> public static void main(String[] args) {</code>
3	<code> int x = 0, y = 0;</code>
4	<code> int sum = 0;</code>
5	<code> double p;</code>
6	<code> while (x < 5) {</code>
7	<code> y = x / 2;</code>
8	<code> while (y < x) {</code>
9	<code> p = (x + 5.0) / 2;</code>
10	<code> sum = (sum + 2) + x + y * 2 + (int) p ;</code>
11	<code> System.out.println(sum) ;</code>
12	<code> y = y + 1;</code>
13	<code> }</code>
14	<code> x = x + 2;</code>
15	<code> if (x > 5) {</code>
16	<code> sum = 2;</code>
17	<code> } else {</code>
18	<code> sum += 3;</code>
19	<code> }</code>
20	<code> }</code>
21	<code> }</code>
22	<code>}</code>

Outputs
12
29
45

Brac University
Department of Computer Science and Engineering

CSE110: PROGRAMMING LANGUAGE I Fall 2025 Quiz 3 Marks: 10

Name:	ID:	Section:
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Question 1: You are a novice programmer tasked with creating a reverse magical character pyramid for a wise wizard. The wizard gives you a single uppercase letter from the English alphabet (A–Z) and asks you to generate a pattern using characters starting from **A**.

However, there is a twist. Instead of building the pyramid upward, you must generate it **in reverse order**.

Wizard's Rules:

- The **first row** must contain all letters from '**A**' **up to the given letter**.
- Each **subsequent row removes the last character** from the previous row.
- The pattern continues until only '**A**' remains in the final row.

Your task is to generate this reverse pattern dynamically based on the input letter. Good luck on this magical quest!

Sample Input	Sample Output
E	ABCDE ABCD ABC AB A
C	ABC AB A
A	A

Tentative Solution:

```
import java.util.Scanner;

public class CharacterPyramid {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter an uppercase character (A-Z): ");

        char input = scanner.next().charAt(0);

        for (char i = input; i >= 'A'; i--) {

            for (char j = 'A'; j <= i; j++) {

                System.out.print(j);

            }

            System.out.println();

        }

    }

}
```

Question 2: Trace the following code and generate the outputs. [Bonus: 3 Marks]

1	<code>public class Q10 {</code>
2	<code> public static void main(String[] args) {</code>
3	<code> int x = 0, y = 0;</code>
4	<code> int sum = 0;</code>
5	<code> double p;</code>
6	<code> while (x < 5) {</code>
7	<code> y = x / 2;</code>
8	<code> while (y < x) {</code>
9	<code> p = (x + 5.0) / 2;</code>
10	<code> sum = (sum + 2) + x + y * 2 + (int) p ;</code>
11	<code> System.out.println(sum) ;</code>
12	<code> y = y + 1;</code>
13	<code> }</code>
14	<code> x = x + 2;</code>
15	<code> if (x > 5) {</code>
16	<code> sum = 3;</code>
17	<code> } else {</code>
18	<code> sum += 2;</code>
19	<code> }</code>
20	<code> }</code>
21	<code> }</code>
22	<code>}</code>

Outputs
11
27
43